Connected device opportunities: where the device is the product

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A third companion report testing where a connected device built on qrwty's dynamic-QR system (one that prints, displays, senses or actuates, is online over Wi-Fi or LTE, and treats every code as a live-controlled redirect with scan logging) can be a viable business, rather than a printed sticker plus any dynamic-QR SaaS.

Companion to the market and business-model assessment and Field device applications. Source numbers are per document: [n] here refers to this report's Sources list, not either earlier report's.

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The first report in this folder found qrwty viable only as a small, open-core business that sells continuity of printed codes, and scored a hardware device as not viable beyond a paid pilot. The second found that for posters, political signs, location hunts and inventory, codes printed in advance and bound to a place by a phone scan make a custom device unnecessary; the founder rejects that answer for a device business, because a pre-printed code plus a phone app is something any dynamic-QR SaaS can offer. This report therefore asks only where a device at the code's location is intrinsic to the job, examines five candidate angles, and costs a small connected fleet against the alternative of selling software for hardware the customer already owns.

The thesis under test. The founder's view is that a QR code backed by an online device is a *control surface*: the server can change what the code does, what the device shows or prints, and can act when a scan arrives, whereas a barcode or pre-printed sticker is inert once it leaves the printer. On this view, the scan log stops being a marketing statistic and becomes an event stream that can drive a lock, a dispenser, a label or a display. This

report treats that as a hypothesis, not as evidence. Each section asks two questions of it: does the device do something a printed dynamic code cannot, and does anyone already pay for that difference?

How to read the evidence labels (the same key as the earlier reports, with two additions): **vendor** = published by the company about itself or its market; **self-reported** = a founder's, seller's or forum member's own figures; **aggregator** = a third-party site that summarises or estimates (Capterra, Vendr, G2, market-research listings and similar); **secondary** = an article relaying another party's figures that this research did not read directly; **retailer** = a shop or reseller listing a product price; **testing vendor** = a certification or test lab describing its own fees; **derived** = arithmetic done in this report on cited numbers; **assumption** = an input with no direct source. Filings, laws and government documents carry no label. Many items in the notes were read only through search-engine summaries, and those are flagged. Prices are USD unless stated, as found on 2026-10-30 and 2026-10-31.

Executive summary

A connected device beats a printed sticker in a short, specific list of jobs: showing a code that changes every few seconds, acting on a scan, printing a label whose identity the server assigns at the moment of an event, and showing content beside the code. Changing where a code points is not on the list, because printed dynamic codes already do it for **$0-29 a month** [1][2]. Across the five angles, the jobs where the device is intrinsic are either owned by chain-scale incumbents (electronic shelf labels, vending payments) or bought as cheap software running on commodity hardware the customer already has (event printing, stamp rallies, attendance, loyalty). The two angles that fit qrwty best are **scan-to-actuate for small unattended operators with tickets above roughly $17-27**, where plain Stripe Checkout undercuts the platform rates found and no self-hostable product turned up in the searches, and **rotating-code presence proof as software that renders on any screen**, where a server-controlled redirect and scan log are the product. Neither has operator demand evidence in the notes. A small fleet's infrastructure costs only about **$0.25-4 per device per month** on the cheaper tools (up to about $8 on balena at 200 devices); support, which no source quantified, is the real cost, and no revenue figure was found for any "bring your own hardware" software business.

Angle	Why a device is intrinsic (or not)	Who pays today, and how much	Incumbents	Gap for a small or self-hostable entrant	Verdict
1 Changeable-QR displays and ESL	Intrinsic only for visual content beside the code and rotating payloads; destination changes need no display [3] [4]	Chains pay $5-20 per label and $80k-110k+ per store (vendor blogs) [5][6]; SMB e-paper signs cost $749 [7]	Vusion (EUR 1,472M FY2025 revenue) [8], Pricer, Hanshow, SoluM; Walmart rolling out to all ~4,600 US stores [9]	Hobby OpenEPaperLink tags at $19-28 with no support [10][11]; no SMB demand evidence found	**Not viable as ESL.** Support OpenEPaperLink tags as a display output for angle 4
2 Scan-to-actuate	Intrinsic: the device is the actuator (lock, relay, dispenser); the code itself can stay a sticker	Vending and laundry operators pay about 3.5-6% per transaction plus $5-10 per device per month [12][13] [14]; doors $30-80 per month (aggregator) [15]	Cantaloupe (1.28M devices) [16], Nayax, PayRange, CSC, Kisi, Noke, Luxer One	No self-hostable "sticker to Stripe to relay" product found; plain Stripe Checkout costs more than incumbents below about $17-27 tickets (derived) [17]	**Conditionally viable niche** for mid-ticket assets without an incumbent; test with one paid pilot
3 Event-driven on-site printing	Intrinsic: the label must be printed when the event happens, with a server-assigned identity	Kitchens about $30 a month (competitor claim) [18]; asset tagging $500-1,000 a year [19]; LOTO software about $1,800 a year (aggregator) [20]	Zebra/Brother hardware plus vertical software (PMI, DateCodeGenie, Brady, Maximo add-ons, UPS Happy Returns) [21][22] [23][24][25]	Small-operator print-on-event with a server record; buyers already own $375-900 printers [26][27]	**Software, not a device.** Drive OEM printers; best niche is asset and work-order labels
4 Presence proof	Intrinsic only for rotating codes, and any screen can show one; outdoors needs a powered station	Japanese organisers pay JPY 40,000 a month to about JPY 700,000 a campaign for QR and GPS software [28] [29]; loyalty SaaS $25-100 a month [30][31]	LY Corporation's LINE mini-app, Decentral, EnjoyTokyo; Loopy, Stamp Me; Eventbrite, Cvent	No weatherproof connected QR station product found; no evidence that organisers pay for presence hardware	**Software feature first** (rotating codes on any screen); station hardware only after a buyer with cash-value incentives asks
5 Fleet plus subscription, and BYOD	Not an angle in itself: the cost structure behind 1-4	Device-plus-subscription prices range from $3.50 per device per month (vendor) [32] to $2,500-3,000 per camera per year	Toast (hardware at a loss) [34], Lime, Flock, Samsara, Vusion	Infrastructure $0.25-4 per device per month on the cheaper tools [35] [36][37][38]; BYOD software has no revenue evidence	**BYOD software first; a reference device kit second; no subsidised fleet**

Angle	Why a device is intrinsic (or not)	Who pays today, and how much	Incumbents	Gap for a small or self-hostable entrant	Verdict
		(Flock, aggregator) [33]			

Three findings carry most of the weight. First, the brief's premise that unattended-payment platforms take 10-15% was not found: the documented rates are **about 3.5-6%**, and on a $2 sale plain Stripe Checkout at 2.9% + 30c costs **17.9%** (derived), so fee arbitrage works only on larger tickets [12][13][17]. Second, in every printing and presence niche examined, the buyer pays for software and treats the hardware as a commodity Zebra, Brother, tablet or phone, which favours selling software for customer-owned hardware over building a device. Third, the cheapest credible device (an ESP32 board with a 4.2-inch e-paper panel in a catalogue box) lands at about **$53-99 a unit at 200 units and $43-77 at 1,000** before support (derived from retail parts and testing-vendor prices, with assumed enclosure costs) [39][40][41]; that is cheap enough for a paid pilot, and too expensive to give away under a $5-10 subscription.

1 What a device adds that a sticker cannot: five capabilities, and destination changes is not one

The five capabilities

A precise list matters because most "smart QR" pitches blur it. A printed dynamic code already lets its owner repoint the destination, count scans and change the landing page at any time; QRStuff sells dynamic-QR menu stands on exactly that promise, updating specials and prices without reprinting cards (vendor) [3], and real-estate rider vendors pitch the same code being repointed to a "sold" page (vendor) [42]. A reusable QR rider costs **$11.00-34.90** [4]. Anything a device offers on top of that has to come from one of five capabilities.

The first is a **rotating or time-bound payload**: the code itself changes every few seconds or minutes, so a photo or screenshot of it stops working. Ticketmaster's SafeTix barcode refreshes about every **15 seconds** so that screenshots and prints are invalid (per a 2024 teardown) [43][44]. A printed code cannot do this by definition, which makes rotating codes the one capability that only a live display can supply. The second is **act-on-scan**: a scan (or a payment confirmed after it) makes something happen at the location, such as a relay opening a lock or starting a machine. Here the code can still be a sticker; the device is the actuator, not the code. The third is **print-on-event with a server-assigned identity**: a label is printed when something happens (evidence is collected, a specimen is drawn, a return is dropped off), and the server decides what the label's code means at that moment. UPS Happy Returns is the cleanest example: the customer brings a QR code, an associate scans it, and the shipping label is printed on site (vendor) [25]. The fourth is **visual content beside the code**: a price, an availability state, a countdown or a "sold" banner that changes with the code. The fifth is **tamper evidence**: a device cannot be overlaid with a fraudulent sticker as easily as a printed code can, which matters

because the FBI has documented criminals placing fake codes over real ones, including more than two dozen on parking stations in Austin [45].

What a device does not add, and what it adds only partly

Destination changes, scan counts, custom domains and continuity are all available without a device. So is location-of-placement capture, which the second report showed a phone can record when a pre-printed code goes up. A device also does not fully solve tampering: a whole device can be swapped or covered, and the official guidance found (FBI, 2022 and 2023) warns people not to scan codes on stickers or codes that look replaced; it does not recommend rotating display codes [45][46]. The tamper argument is a pitch, not a standard. Finally, rotating codes stop screenshot sharing but not a person relaying a live image to a friend over a video call; stronger presence proof needs a second factor such as device binding or location, which is what a 2023 peer-reviewed attendance system combined with dynamic QR (academic) [47].

A device adds capabilities; changing the destination is not one of them

Capability by option: yes, partly or no (derived assessment from the cited capabilities); as of 2026-10-31

Yes (1) Partly (0.5) No (0)

Capability	Static printed QR	Printed dynamic QR + SaaS	Code rendered on user's phone (SafeTix-style)	Connected display at location	Connected actuator (sticker code)	Connected printer at location
Change destination after printing	no	yes	yes	yes	yes	yes
Scan logging	no	yes	yes	yes	yes	yes
Rotating or time-bound payload at the location	no	no	partly on the user's device, not at the location	yes	no	no
Act on scan at the location	no	no	no	no	yes	no
Print with server-assigned identity at event time	no	no	no	no	no	yes
Visual content beside the code	no	no	no	yes	no	partly on the label, once
Resists sticker overlay	no	no	yes	partly device can be swapped	no	partly

Source: vendor and news sources cited in the report (section 1); scores are the report's assessment; see data/device-vs-sticker-capability-matrix.csv.

▼ Data

Capability	Static printed QR	Printed dynamic QR + SaaS	Code rendered on user's phone (SafeTix-style)	Connected display at location	Connected actuator (sticker code)	Connected printer at location	Source
Change destination after printing	0	1	1	1	1	1	[3][42]
Scan logging	0	1	1	1	1	1	[1][2]
Rotating or time-bound payload at the location	0	0	0.5 (on the user's device, not at the location)	1	0	0	[43]
Act on scan at the location	0	0	0	0	1	0	[16] [48]
Print with server-assigned identity at event time	0	0	0	0	0	1	[25]
Visual content beside the code	0	0	0	1	0	0.5 (on the label, once)	[49]
Resists sticker overlay	0	0	1	0.5 (device can be swapped)	0	0.5	[45]

Scores: 1 = yes, 0.5 = partly, 0 = no (derived assessment from the cited capabilities).

2 Changeable-QR displays and ESL: a $2 billion market where QR is a side feature

Chain-scale vendors and Walmart define the market

Electronic shelf labels are the obvious commercial form of "a display that shows a code". Analyst estimates of the 2025 market range from **$1.61B to $2.66B**, with growth rates of 11-21% a year; Global Market Insights puts 2025 at $2.2B and 335.9M units (aggregator) [50][51][52][53]. The figures come from syndicated listings with differing methods, so the range is the honest answer. The leader's numbers are firmer. VusionGroup reported FY2025 IFRS revenue of **EUR 1,472M (+54%)** and adjusted revenue of about EUR 1,527M (+51%), with adjusted EBITDA of EUR 277M (18.2% margin), in a press release issued while its audit was still in progress (vendor) [8]. Pricer, a smaller listed rival, had net sales of SEK 2,147.9M, down from SEK 2,558.0M, at a **22.2% gross margin**, with order intake down from SEK 2,918.4M to SEK 1,892.1M (vendor) [54].

Walmart is the demand engine. It signed a first phase for 500 stores and about 60M labels in 2023, expanded to 2,300 stores by 2026 in June 2024, and in December 2024 extended the programme to all of its roughly 4,600 US stores; one report values the order at about EUR 1bn, and store counts vary by report (vendor, trade press) [55][9]. Walmart Mexico is

installing more than 1.7M labels with completion in 2026 [56]. The Walmart figures imply about **120,000 labels per store** (derived: 60M / 500), an order of magnitude above the ~10,000-label store used in reseller cost breakdowns; the two describe different store formats rather than contradicting each other.

Hardware dominates revenue, and per-label recurring revenue is not disclosed

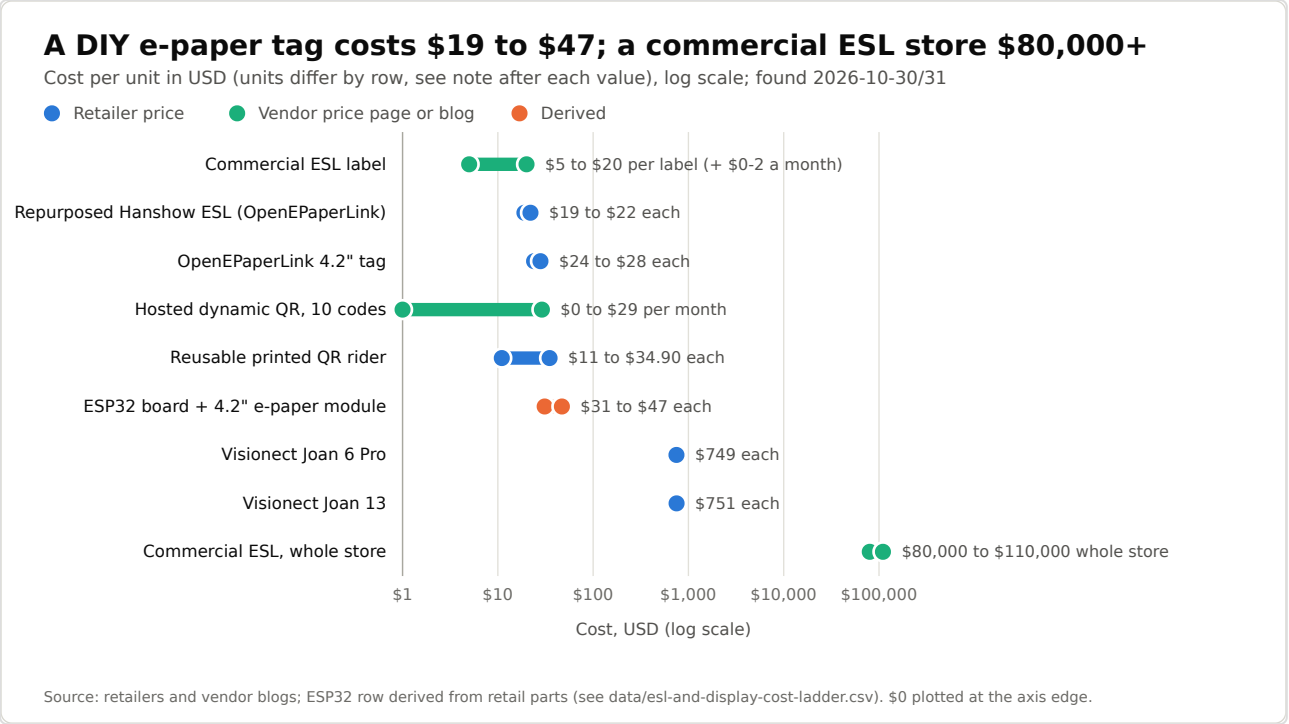
Labels cost roughly **$5-20 each** (small black-and-white $5-10, colour $10-20), and a store with about 10,000 labels runs to **$80,000-110,000 or more**; one breakdown for a supermarket lists labels at $150k-450k, rails and network $15k-50k, installation $20k-60k and software $10k-40k a year, and some vendors add $0-2 per label per month in cloud fees. These are vendor and reseller blogs, some dated 2020-22 [5][6][57]. Vusion's software line is growing but small: value-added services were about **EUR 211M**, roughly 14% of revenue, and recurring value-added services were **EUR 83M (+45%)**, passing EUR 100M annualised in Q4 2025 (vendor) [8]. Recurring revenue is therefore about **5.6% of IFRS revenue** (derived: 83 / 1,472). Vusion guides to 15-20% growth in 2026, including about 40% growth in value-added services [8]. No source disclosed revenue per label per year (no evidence found).

QR codes are a secondary feature in this market. Hanshow says its labels integrate with product-information systems and GS1 Digital Link for Digital Product Passport access, and other vendors describe QR or NFC on labels for product details or loyalty offers (vendor) [49][58]. No source confirmed Walmart using QR on its shelf labels (no evidence found). Buyers pay for price automation; a QR redirect or scan log would be an add-on to someone else's label, sold into a procurement process designed for chains.

The SMB display gap is a price gap with no demand evidence

Below the chains, dedicated e-paper signage is expensive. Visionect's Joan 6 Pro lists at **$749** and the Joan 13 at $751 at one reseller; these are room-booking displays, not QR products [7][59]. The only "e-ink QR" table stand found was an AliExpress wiki tag with no specifications or price [60]. At the other end, the open-source OpenEPaperLink ecosystem sells preflashed 4.2-inch tags for **$24 (one or two units) or $28 (three or more)** and repurposed Hanshow labels for $19-22, as-is with no warranty; they need an ESP32 access point, and some tags need opening or drilling to flash (retailer, self-reported) [10][11][61]. One seller says it no longer sources displays. Building from modules, a Waveshare ESP32 e-paper driver board costs **$12.33 at one unit and $11.56 at 100 or more**, and 4.2-inch modules cost **$18.99-34.99** (retailer, vendor) [39][40], so a bare electronics kit is about **$31-47** before case, battery or certification (derived).

That leaves a plausible gap between $30 hobby tags and $749 room signs, perhaps a hosted e-paper QR sign at $30-80. The notes found no demand for it: no SMB asking for powered changeable-QR displays, and no SMB-priced vendor selling Wi-Fi e-paper QR signs (no evidence found). The only display capability a printed dynamic code cannot copy is the rotating payload, and that belongs to presence proof (section 5).



▼ **Data**

Option	Unit cost (USD)	Basis	Source
Hosted dynamic QR, 10 codes	0-29 per month	vendor price pages	[1][2]
Reusable printed QR rider	11.00-34.90	retailer	[4]
Repurposed Hanshow ESL for OpenEPaperLink	19-22	retailer, no warranty	[11]
OpenEPaperLink 4.2" tag	24-28	retailer, access point extra	[10]
ESP32 driver board + 4.2" e-paper module	31-47	derived from retail parts (11.56-12.33 + 18.99-34.99)	[39] [40]
Commercial ESL label	5-20 per label, plus 0-2 per label per month cloud fee	vendor blogs	[5][6]
Commercial ESL, whole store (~10,000 labels)	80,000-110,000+	vendor blogs	[5][57]
Visionect Joan 6 Pro	749	retailer	[7]
Visionect Joan 13	751	retailer	[59]

Verdict. Do not enter ESL. The market is real but bought by chains from vendors with billion-euro revenue, and QR is incidental to it. The useful part for qrwty is technical: OpenEPaperLink tags are a $20-30 display that customers can buy themselves, which makes them a candidate output for rotating codes.

3 Scan-to-actuate for small unattended operators: incumbents take about 4-6%, not 10-15%

The model is proven at scale, as a payments business

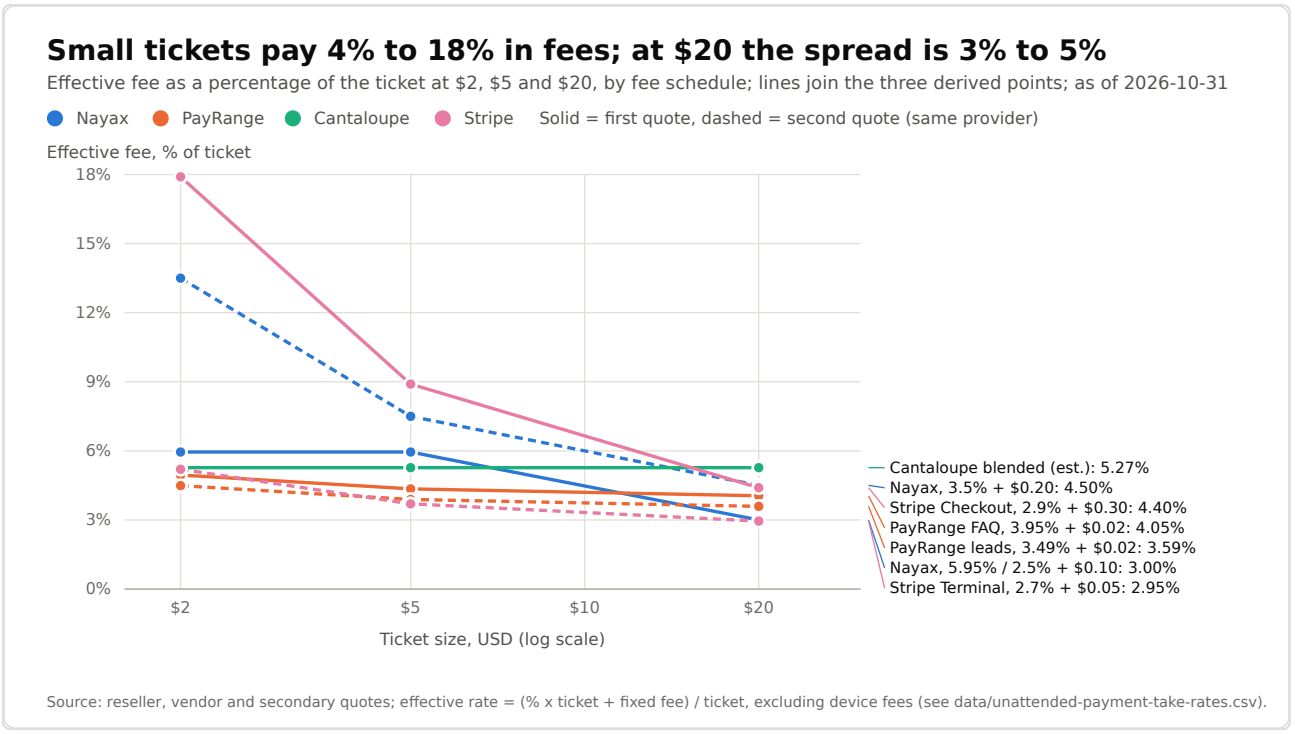
The pattern "scan, pay, the machine acts" is established in vending, laundry, EV charging, micromobility and car washes. Cantaloupe's FY2025 10-K reports **1.28M active devices** and **34,896 active customers**; most customers pay a monthly service fee plus a blended percentage of transaction volume, equipment is sold, financed or bundled into a monthly "Cantaloupe ONE" subscription, and agreements can generally be cancelled on 30 days' notice [16]. Third-party summaries put FY2025 revenue at about $303M and transaction volume at $3.4B (secondary; not verified from the filing text). That works out to about **$237 of revenue per device per year, or roughly $20 a month**, across subscription, processing and equipment (derived), and an average of about 37 devices per customer (derived). An investor newsletter estimates Cantaloupe's take rate at about **5.27%** in Q1 FY2025, up from 4.84% in FY2022 (secondary) [14].

Lime shows the economics when the device itself earns money. Trade coverage of its May 2026 S-1 reports FY2025 revenue of **$886M** (one passage says $887M), adjusted EBITDA of $218.1M, adjusted gross margin of about 53%, 325,137 vehicles, **$7.47 of revenue per vehicle per day**, and a fully landed vehicle cost of about $1,300 with about a 12-month payback (secondary) [62]. Lime is not a small-operator model, but the arithmetic is the one a qrwty actuator has to beat: the device must either earn transaction revenue at the location or save an operator enough to justify a subscription.

The documented take rates are 3.5-6%

The brief assumed that small operators pay 10-15% to platforms. No source in the notes supports that for any hardware-payment incumbent. Nayax resellers quote **5.95% on sales of $5 or less and 2.5% + $0.10 above $5**, other resellers quote 3.5% + $0.20 or 3.0% + $0.20, and cashless plans cost about $7.95-9.99 per device per month with a $30 activation at some resellers; no official Nayax rate card was found (retailer) [12][63]. PayRange's FAQ lists **3.95% in the US plus a $0.02 fee**, 5.25% for amusement machines, and a $79.95 BluKey; a separate PayRange leads page says 3.49% + $0.02 and $59.95 plus a $20 bundle, and the two pages conflict (vendor) [13][64]. PayRange cites an industry average of 6-7% and Crane advertising 5.4% (vendor) [13]. The highest figure found anywhere is that 6-7% industry claim; the 10-15% premise should be dropped. Table-ordering fees from me&u appear in customer reviews at about 1.05-1.6% (aggregator) [65].

The comparison that matters for a self-hosted entrant is against plain Stripe. Stripe's standard online price is **2.9% + 30c**, with Checkout and Payment Links included; in-person Terminal pricing is 2.7% + 5c but needs a $59-329 reader (vendor) [17]. The fixed 30c dominates small tickets.



▼ **Data**

Fee schedule	Quoted rate	Effective at $2	Effective at $5	Effective at $20	Source
Nayax (reseller quote)	5.95% at $5 or less; 2.5% + $0.10 above	5.95%	5.95%	3.00%	[12]
Nayax (alternative reseller quote)	3.5% + $0.20	13.50%	7.50%	4.50%	[12]
PayRange (FAQ)	3.95% + $0.02	4.95%	4.35%	4.05%	[13]
PayRange (leads page)	3.49% + $0.02	4.49%	3.89%	3.59%	[64]
Cantaloupe blended take rate	~5.27% (estimate)	5.27%	5.27%	5.27%	[14] (secondary)
Stripe online (Checkout, Payment Links)	2.9% + $0.30	17.90%	8.90%	4.40%	[17]
Stripe Terminal (needs $59-329 reader)	2.7% + $0.05	5.20%	3.70%	2.95%	[17]

Effective rates are derived: (percentage x ticket + fixed fee) / ticket. They exclude monthly device fees and any qrwty fee.

The break-even points are derived from the same quotes. Stripe Checkout matches PayRange's 3.95% + $0.02 at a **$26.67 ticket** and Nayax's alternative 3.5% + $0.20 quote at **$16.67**; it never beats Nayax's quoted 2.5% + $0.10 above $5. Below those levels a QR-to-Stripe design costs the operator more than the incumbent, before qrwty charges anything. Laundry machines, snack vending and scooter unlocks (about $1 plus $0.15-0.52 a minute, secondary) [66] are small-ticket. EV charging sessions, equipment and tool rental, campsite or marina amenities, and makerspace machine time plausibly sit above the line, but the notes contain no ticket-size or operator data for them (no evidence found).

What operators complain about, and where nothing was found

Operator evidence is thin. A laundromat owner quoted in trade press says "transaction fees definitely are real, but customer convenience is king" and that app users seem to spend more; another calls card processing fees "ridiculous"; the same feature puts card processing at 1.5-3% of a sale (trade press, search summary) [67]. An RV park operator on a forum says laundry sales dropped sharply after switching to app payment (self-reported) [68], and one consumer review calls a laundry app's pay step a way to save the owner card fees (aggregator) [69]. Beyond these two owner quotes and two anecdotes, no operator fee or lock-in complaints were found for car washes, campgrounds, marinas, tool libraries or makerspaces (no evidence found). Cantaloupe's 30-day cancellation terms suggest switching is not hard [16].

Access and lockers are monetised per door or per unit rather than per scan. Vendr estimates Kisi door access at **$30-80 per door per month** depending on size, with a median annual spend of about $5,100-5,250, figures it reports inconsistently (aggregator) [15]; Kisi's own blog cites $600-1,600 a year per door (vendor) [70]. Noke smart entry for self-storage costs "roughly $200 to $300 per smart lock", including installation and the tenant app [71], and a Janus study says operators can charge tenants $5 a month for after-hours or $15 a month for 24-hour access (vendor, search summary) [72]. A search snippet attributed $13.90 per lock and a $20 membership to the same Noke article; the fetched page showed those figures belong to a different vendor, so they are excluded [71]. Tenant Turner claims its devices cost $3.50 per device per month against $5 for others (vendor) [32]. Parcel lockers are quote-based at $6,000-20,000 plus a monthly fee [73].

Self-hostable options exist only at the edges

The searches found open or cheap back-ends in only one category. In EV charging, OpenCPO is an Apache 2.0, OCPP 1.6/2.0.1 back-end with a QR-start web app (aggregator) [74], askacharge is listed at EUR 20 a month per brand with 0% commission and app-free QR (aggregator) [75], EV Connect launched app-free QR charging (vendor) [48], the Open Charge Alliance published a June 2026 whitepaper on QR payments over OCPP [76], and ChargePoint announced ad hoc QR payments with Cariqa in August 2026 [77]. EV charging is therefore already served at the low end. Outside EV, the only self-hostable QR lock found was a hobby project that runs a camera scanner on a Raspberry Pi with a 12 V relay (self-reported) [78], a different architecture in which the device reads a phone instead of a phone reading a sticker. No ready, self-hostable "scan sticker, pay through Stripe, signed webhook, relay" product was found; that is consistent with a gap but does not prove one.

The hardware for the actuator is cheap. KinCony sells ESP32 relay boards with a 4G modem from **$40** (two channels) to $140 (vendor) [79]; SIMCom Cat-1bis modules cost about $18-20 at one unit (retailer) [80]; Hologram charges **$1 per SIM per month plus $0.03 per MB** and $3 per SIM card (vendor) [38]. A scan-and-unlock event is a few kilobytes, so data costs cents a month and the SIM fee dominates (derived). Enclosure, power supply and lock costs were not researched.

What SAQ A does and does not mean (inference to confirm)

The architecture keeps card data away from qrwty. Stripe says low-risk integrations send payment information "directly to Stripe without it passing through your servers, reducing your PCI obligations", that PCI is a shared responsibility, and that the business must still "annually attest" (vendor) [81]. Stripe's PCI guide says Checkout and Elements host card inputs in a Stripe-domain iframe, which meets "PCI eligibility for the SAQ A"; it does not mention Payment Links (vendor) [82]. Third-party blogs describe redirect-to-Checkout as SAQ A and Elements variously as SAQ A or SAQ A-EP (vendor) [83].

The inference, which the operator should confirm with its acquirer and against the PCI Security Standards Council's current SAQ A text (not read in this research), is that a static code leading to Stripe Checkout, with a verified webhook commanding the device, leaves the merchant in the SAQ A tier. SAQ A does not mean no obligations: the merchant still attests each year, the page that links to checkout must be served securely, and Stripe asks for webhook signature checks and TLS 1.2 or later [81]. One vendor blog claims PCI DSS 4.0 requirements 6.4.3 and 11.6.1 (script management and change detection on payment pages) apply to hosted-checkout merchants that load scripts; that claim could not be confirmed against PCI SSC text and conflicts with the reading that SAQ A merchants host no payment page [84]. SAQ A also says nothing about the risks that matter most here, which are product design issues: replayed webhooks, double vends, payment taken but machine failed, refunds, and accessibility. One accessibility vendor says no law requires QR codes but that QR flows fall under accessibility obligations once used, and the kiosk industry association has a 14-point ADA checklist; no source said QR-only access is prohibited (vendor) [85][86]. A keypad, staff call or card fallback is prudent (assumption, not legal advice).

Verdict. Conditionally viable as a niche: a self-hostable scan-to-actuate kit for operators of unattended assets with tickets above roughly $17-27 and no incumbent platform, sold on ownership and per-scan logs rather than on fees. Not viable for small-ticket vending or laundry, where incumbents are cheaper than plain Stripe. Operator demand is unproven.

4 Event-driven on-site printing niches: printing is mandatory, the printer is a commodity

Nine niches compared

This angle has the strongest claim that a device at the location is intrinsic: some labels cannot exist until an event happens. The notes examined nine niches; evidence for all of them comes from search summaries without page fetches.

Niche	Mandatory on-site printing?	Incumbents and prices	Would a dynamic redirect add value?	SMB or self-host gap
Evidence (police property rooms)	Yes: the label must exist at collection. Suffolk County, NY prints 70,000+ evidence labels and invoices a year on one Zebra (vendor) [87]	PMI Evidence Tracker bundles software, printer, scanner, labels and a year of support, no price (vendor) [21]; Evidence Hound prints via Teklynx (self-reported) [88]	Low: the code must resolve to an auditable record; a static ID is safer (derived judgement)	Agency site licences; no SMB or self-hosted option found
Clinical specimens	Yes: labelled at the bedside on Zebra healthcare printers (retailer) [89]	Zebra ZQ320 Plus about $375, QLn320 HC $570-790 (retailer) [26]; ZQ610/620 Plus HC GBP 687-896 ex VAT (retailer) [27]	Low, for the same audit reason	Hospital systems; no gap evidence
Field service and work orders	Often: Accuviti and Opqo print asset labels on site from Maximo records (vendor) [90][24]; an IBM Ideas request suggests network printing is not native (self-reported) [91]	GoCodes $500 a year for 200 assets, $1,000 Premium (vendor) [19]; QR labels under $0.15 each (vendor) [92]; Fiix gives each asset and work order a QR code (vendor) [93]	**High**: the asset's code can resolve to its current work order or status	Maximo add-ons are enterprise-only [24]; small maintenance teams are the clearest gap
Construction and safety tags (LOTO, scaffold)	Partly: LOTO procedures sit behind QR codes printed on a Brady printer (vendor) [23]; scaffold tags are pre-made packs at C$42-48 (retailer) [94]	Brady LINK360 about $1,800 a year (aggregator) [20]; Brady printers $1,500-4,000 (aggregator) [95]	**High** for LOTO: the code points to the current procedure	Brady is the incumbent; no SMB price found
Regulated cannabis inventory (Metrc)	Tags are state-issued, assigned in the field	Plant tags $0.46 and package tags $0.26 in Michigan from 1 October 2025 [96]; Massachusetts the same [97]; Montana $0.55 and $0.32 [98]	Low: the official label's form is fixed by the state	A qrwty code could only be an extra sticker
Kitchen date labels	Yes: labels per batch and day	DateCodeGenie printer plus app subscription (vendor) [22]; about $30 a month and up to $2,000 setup (competitor claim) [18]; Lite system GBP 1,248.97 ex VAT (retailer) [99]; $0.06 per label (retailer) [100]	Medium at best: a recall or allergen page is plausible but unevidenced	Crowded; no Toast or Square label pricing found
Returns drop-off	Yes: UPS Happy Returns prints the label at drop-off and ordinary drop boxes cannot process QR returns (vendor) [25] [101]	UPS and its retail network (The UPS Store, Ulta, Staples) [101]	Already a pointer to a server record	Carrier-scale; no small-entrant gap
Citations and event badges	Citations yes; badges optional, since Eventbrite says on-site printing is unnecessary when attendees pre-register (vendor) [102]	Passport uses Bluetooth Zebra printers (vendor) [103]; Santa Ana bundles a Zebra ZQ511, data plan, licence and management per unit [104]; 2017 bid $300 per unit per year or $50 a month lease [105]; badges on Brother	Low for citations (legal record); medium for badges	Municipal procurement; event tools already print badges

Niche	Mandatory on-site printing?	Incumbents and prices	Would a dynamic redirect add value?	SMB or self-host gap
		QL-820NWB (vendor) [106]; Cvent badge app costs extra (competitor claim) [107]		
EU plant passports	No: content and form set by Regulation 2017/2313 and can be batch-printed; records kept 3 years [108] [109]; may be held digitally (UK) [110]	Authorised operators' own systems	Low: fixed regulated format	Add-on sticker only

Where the value sits

Two patterns run through the table. The printer is a commodity in every niche, a $375-900 Zebra or a low-cost Brother, and the value sits in the integration with a record system, which is the "sticker plus SaaS" pattern the founder wants to avoid, except that the sticker is printed at the event rather than in advance [26][27]. And the dynamic redirect, qrwty's core feature, matters least where on-site printing matters most: evidence, specimens and citations need codes that resolve to a fixed, auditable record, and Metrc tags and EU passports have state-fixed content [96][108]. What qrwty can still add there is *server-assigned identity*: issuing a unique ID at print time and logging every scan, which a static barcode cannot log.

The overlap of "must print on site" and "a live redirect helps" is narrow: field assets and work orders, and LOTO procedures. Prices there are **$500-1,800 a year** for software [19] [20], with label media under $0.15 [92]. The indicated per-device benchmark across kitchens and citation handhelds is about **$30-50 a month** (competitor claim and a 2017 bid) [18][105]. Buyers already own the printers or buy them from Zebra and Brother resellers.

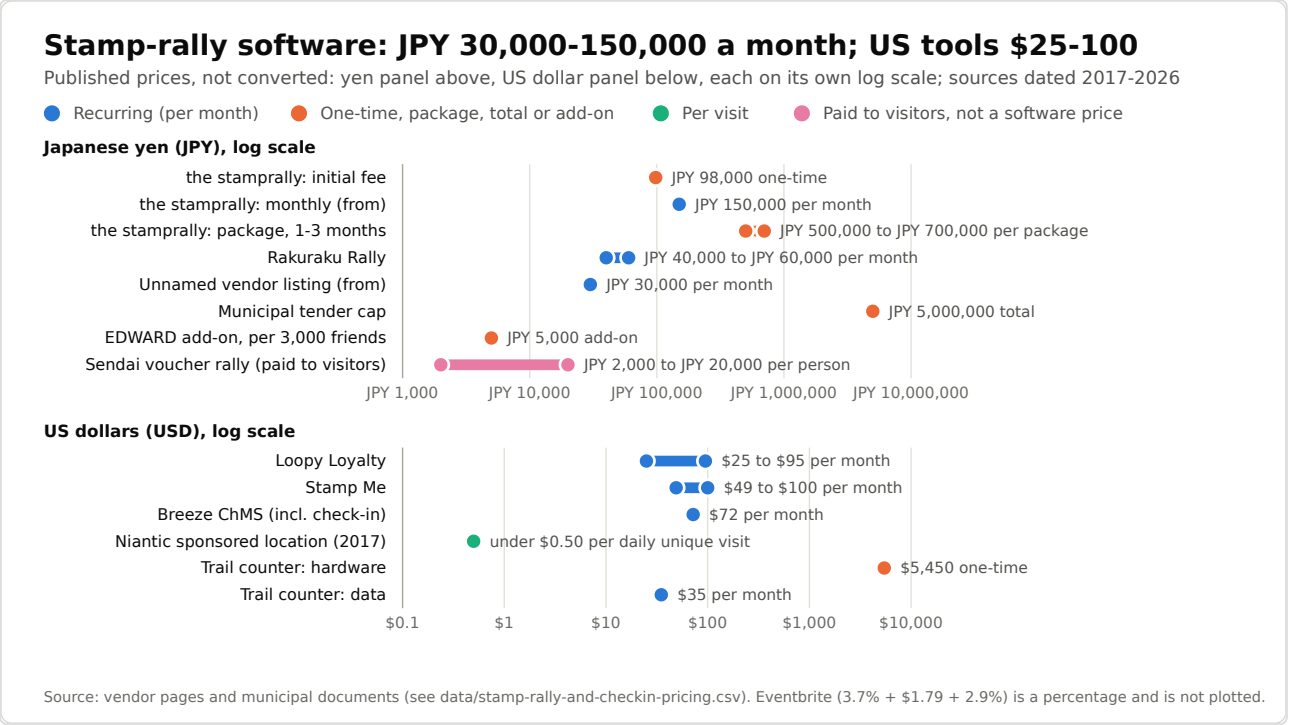
Verdict. The device is intrinsic but should not be built. A print-on-event feature that drives customer-owned Zebra and Brother printers, issues server-assigned codes and attaches scans to a work order or procedure is a software product with a clear niche (small maintenance and safety teams) and a priced benchmark. It is a better fit for qrwty than kitchens, returns or regulated labels.

5 Presence proof: stations, stamp rallies, loyalty, attendance (buyers pay for cheap software, not hardware)

Japanese digital stamp rallies are cheap QR and GPS software projects

Japan has the most developed organiser-pays market for "prove you visited" campaigns, run by municipalities, tourism boards and shopping streets. Its prices are published, and

they are for software. All amounts below are in yen, as stated by the source on the date given.



Source: vendor pages and municipal documents (see data/stamp-rally-and-checkin-pricing.csv). Eventbrite (3.7% + $1.79 + 2.9%) is a percentage and is not plotted.

▼ Data

Product or programme	Price	Date of source	Methods	Source
the stamprally (LY Corporation, LINE mini app)	JPY 98,000 initial + from JPY 150,000 per month; about JPY 500,000 (1 month, 10 spots, 1,000 people), JPY 600,000 (2 months, 30 spots, 2,000), JPY 700,000 (3 months, 50 spots, 3,000), tax incl.	undated page, fetched 2026-10-31	QR, GPS, on-screen stamp, LINE touch, riddle; NFC not mentioned	[29]
Rakuraku Rally (Decentral, Kyoto)	JPY 40,000 per month (31+ days); JPY 60,000 spot plan (30 days or less); no initial fee listed	2026-05-29	QR, GPS, keyword	[28]
Unnamed vendor listing	JPY 0 initial, from JPY 30,000 per month	undated (attribution uncertain)	not stated	[111]
Municipal tender cap	JPY 5,000,000 total; GPS not required, stamps by QR	date not confirmed	QR	[111]
EDWARD (LINE mini app)	+JPY 5,000 ex tax per extra 3,000 friends; base fees not captured	undated (search summary)	not stated	[112]
Sendai "Machi-iko" voucher rally	4 stamps earn JPY 2,000 in vouchers, up to JPY 20,000 per person; JPY 2.1bn issuance budget; 1,826 stores	2022-10-18	QR	[113]
Loopy Loyalty	USD 25 / 69 / 95 per month	accessed 2026-10-31	not stated	[30]
Stamp Me	USD 49 (limited-time) / 79 / 100 per month	accessed 2026-10-31	single-use codes, tap tags	[31]
Breeze ChMS (includes check-in)	USD 72 per month	accessed 2026-10-31	not stated	[114]
Eventbrite paid tickets	3.7% + USD 1.79 per ticket + 2.9% processing	accessed 2026-10-31	not applicable	[115]
Niantic sponsored location	under USD 0.50 per daily unique visit	2017-05-31	GPS within 40-50 m	[116] [117]
Trail counter (Urban Post Multi)	USD 5,450 one-time + USD 35 per month data	2021 (search summary)	infrared counter	[118]

Municipal specifications confirm that QR and GPS are the norm. Yokohama's proposal materials list QR, quiz and GPS as options [119]; Kobe grants a GPS stamp within 50 m of a point (2026-08-04) [120]; Fukuroi's 2026 specification makes QR scanning the base method with GPS at tourist facilities (search summary) [121]; Kagoshima uses QR indoors and GPS for outdoor statues [122]. One vendor combines GPS and QR, checking location at the scan, to prevent fraud (vendor) [123]. None of the platforms found mentions NFC, and no presence-proof hardware appears in any specification (no evidence found).

Campaigns are small. Gifu's industrial-tourism rally had **5,303, 4,077 and 7,440 participants** in three successive years [124]; a Nagano water-story rally had 121 against a target of 50 [125]; Kyoto's Nishijin rally had 429 [126]; Kanagawa distributed 256 NFT stamps with 30 completers [127]. Software for such a campaign costs roughly JPY 40,000 a month to JPY 700,000 in total [28][29], and Shizuoka's subsidy rules list prizes, production and honoraria as eligible costs [128], so the buyer is buying a campaign, not equipment. A station-hardware add-on would have to fit inside that budget.

Where rotating codes matter: when the stamp is worth money

Cheating with static codes matters where a stamp converts to cash. Sendai's rally paid **JPY 2,000 in vouchers per four stamps, up to JPY 20,000 per person, from a JPY 2.1bn issuance budget** across 1,826 stores [113]; at that value, a photographed code shared on social media is a direct cost. The largest money flow in these programmes is the incentive, not the software fee. GPS alone is defeatable: an Aalborg thesis documents Pokemon GO players spoofing location (search summary) [129], a fan-site poll suggests 20-25% spoof, though the poll's own authors flag it as unreliable (self-reported) [130], and spoofing with software-defined radios is documented [131]. A rotating code at the venue raises the bar against screenshot sharing; combined with a location or device check it approaches the design a 2023 peer-reviewed attendance study reported as blocking the attacks it tested (academic) [47]. No source showed a tourism organiser paying for rotating codes (no evidence found).

The other paying analogue is the verified visit. Niantic charged sponsors **under $0.50 per daily unique visit** in 2017, counting only players within 40-50 m as an engaged visit; the programme page is now marked temporarily suspended and US-only, with no public prices [132][116][117]. That supports a price unit of cents per verified visit, but only at Niantic's scale.

Indoors any screen works; outdoors nothing is on sale

A rotating code needs a live display, but not a custom one. Indoors, a tablet or old phone running a browser page can show a code that the server rotates; that is a software product on customer-owned hardware. Outdoors, the notes found no product sold as a weatherproof connected QR check-in station, in English or Japanese searches (no evidence found). The nearest analogues are trail counters, which public agencies do buy: an Eco-Counter with Pyro-Box at about $4,000 plus $550 per extra counter (2017, search summary, unverified) [133], and an Urban Post Multi at **$5,450 plus $35 a month** for automatic data transmission (2021, search summary) [118]; current Eco-Counter units claim batteries lasting up to two years (vendor) [134]. Estimote's 2018 LTE beacon cost about $129 plus about $2 a month in data (search summary) [135]. The vandal-resistant QR hardware found reads codes at gates rather than showing them (vendor) [136]. A solar, rotating-code station under about $500 would undercut counters roughly tenfold (derived from the counter prices), but counters sell traffic data to agencies that need it for grants and planning, not presence proof to visitors, and no buyer for the latter was found.

Venue loyalty and attendance follow the same pattern. Loopy Loyalty charges **$25-95 a month** [30] and Stamp Me **$49-100**, already offering single-use alphanumeric or QR "Onestamps" and tap tags, which is presence proof in software [31]. Planning Center Check-Ins has a free tier for 10 daily check-ins [137], Breeze costs $72 a month with check-in included [114], and Cvent's OnArrival is quote-only, with an annual licence plus a per-registrant fee [138].

Verdict. The market pays for cheap QR and GPS software, not presence hardware. Rotating codes matter where a stamp, visit or check-in is worth money (voucher rallies, prize draws, paid attendance), and qrwty can serve that as a software feature: per-scan

token validation, a short time window, an optional location check, and a scan log that is the proof. Dedicated station hardware is justified only when an organiser with cash-value incentives needs outdoor points with no power or screen.

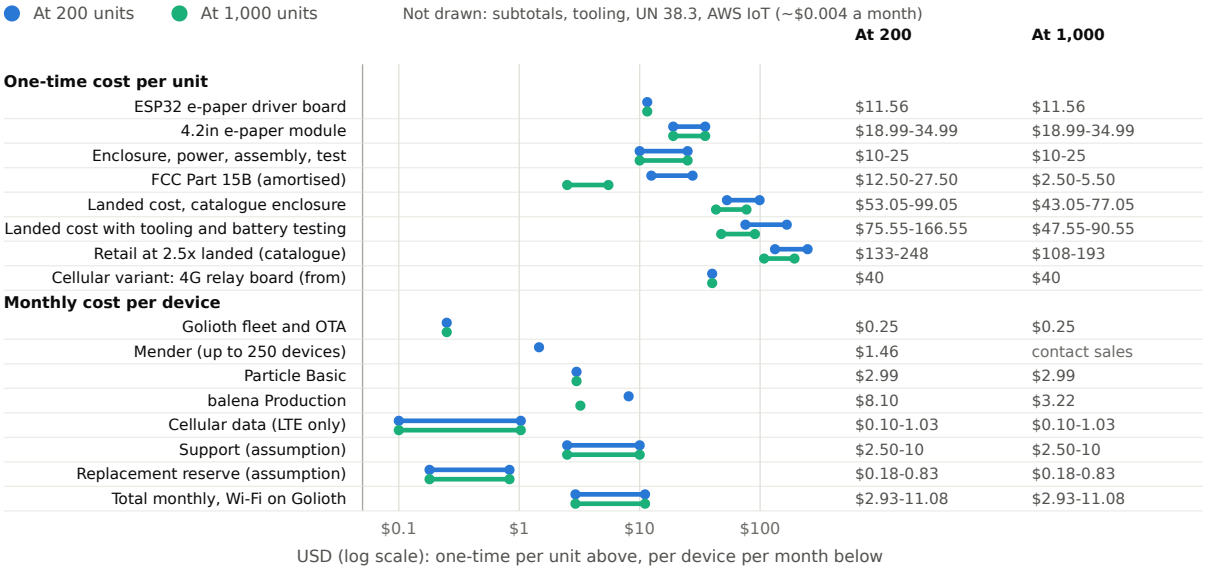
6 Fleet economics: build, certify, connect, update, support (infrastructure is cents; support is the cost)

Cost per unit at 200 and 1,000 units

No published bill of materials exists for a comparable product, so the table builds one from cited parts and labels every unsourced line. The reference device is the cheapest credible station: an ESP32 e-paper driver board and a 4.2-inch e-paper panel on Wi-Fi in a catalogue enclosure, the same core that serves a display, a rotating-code station or (with a relay) an actuator. Part prices are retail at one to 100 units because no volume quotes were found; real 1,000-unit costs would be lower by an unknown amount.

At 1,000 units landed cost falls about a fifth; monthly cost does not

USD per unit one-time (above) and per device per month (below); low-high range where listed; 2026-10-31



	At 200	At 1,000
One-time cost per unit		
ESP32 e-paper driver board	$11.56	$11.56
4.2in e-paper module	$18.99-34.99	$18.99-34.99
Enclosure, power, assembly, test	$10-25	$10-25
FCC Part 15B (amortised)	$12.50-27.50	$2.50-5.50
Landed cost, catalogue enclosure	$53.05-99.05	$43.05-77.05
Landed cost with tooling and battery testing	$75.55-166.55	$47.55-90.55
Retail at 2.5x landed (catalogue)	$133-248	$108-193
Cellular variant: 4G relay board (from)	$40	$40
Monthly cost per device		
Golioth fleet and OTA	$0.25	$0.25
Mender (up to 250 devices)	$1.46	contact sales
Particle Basic	$2.99	$2.99
balena Production	$8.10	$3.22
Cellular data (LTE only)	$0.10-1.03	$0.10-1.03
Support (assumption)	$2.50-10	$2.50-10
Replacement reserve (assumption)	$0.18-0.83	$0.18-0.83
Total monthly, Wi-Fi on Golioth	$2.93-11.08	$2.93-11.08

Source: retailer and vendor pages; amortised, support and reserve rows derived or assumed (see data/fleet-cost-per-unit-200-vs-1000.csv).

▼ Data

Line	At 200 units (USD per unit)	At 1,000 units (USD per unit)	Basis	Source
ESP32 e-paper driver board	11.56	11.56	retailer, 100+ price	[39]
4.2" e-paper module	18.99-34.99	18.99-34.99	vendor, single-unit price	[40]
Catalogue enclosure, power, assembly, test	10-25	10-25	assumption (no IP65 enclosure price found)	none
Parts subtotal	**40.55-71.55**	**40.55-71.55**	derived	
FCC Part 15B with pre-certified module ($2,500-5,500 per model)	12.50-27.50	2.50-5.50	testing vendor, amortised (derived)	[41][139]
Landed cost, catalogue enclosure	**53.05-99.05**	**43.05-77.05**	derived	
Optional: injection-mould tooling ($4,000-12,000)	20-60	4-12	vendor, amortised (derived)	[140] [141]
Optional: UN 38.3 for a lithium pack ($500-1,500)	2.50-7.50	0.50-1.50	testing vendor, amortised (derived)	[142] [143]
Landed cost, with tooling and battery testing	**75.55-166.55**	**47.55-90.55**	derived	
Retail at 2.5x landed cost (catalogue enclosure)	133-248	108-193	derived from the rule of thumb	[144]
Cellular variant: ESP32 4G relay board instead of driver board	from 40	from 40	vendor	[79]
Monthly, per device				
Fleet and OTA: Golioth	0.25	0.25	vendor	[35]
Fleet and OTA: Mender Professional ($291 per month up to 250)	1.46	contact sales above 250 tier	vendor (derived per device)	[145]
Fleet and OTA: Particle Basic ($299 per 100-device block)	2.99	2.99	vendor	[36]
Fleet and OTA: balena Production ($1,439 for 110, +$2 each)	8.10	3.22	vendor (derived per device)	[146]
Message broker: AWS IoT Core connectivity	~0.004	~0.004	vendor ($0.042 per device-year)	[147]
Cellular data (LTE variant only)	0.10-1.03	0.10-1.03	vendor: 1NCE EUR 12-USD 15 per 10 years; Hologram $1 + $0.03 per MB at 1 MB	[37][148] [38]
Support: 0.5-2 hours per device per year at $60 an hour	2.50-10.00	2.50-10.00	assumption (no benchmark found)	none
Replacement reserve: 5-10% of units a year	0.18-0.83	0.18-0.83	assumption (no RMA rate found)	none
Total monthly, Wi-Fi on Golioth	**2.93-11.08**	**2.93-11.08**	derived	

At 200 devices, a $5-10 subscription brings in **$12,000-24,000 a year** (derived), of which Golioth infrastructure is about $600 (derived: $50 a month). Under the support assumption, the same fleet needs **100-400 hours of founder time a year**; at 1,000

devices it is 500-2,000 hours, up to a full-time job (derived from the assumption). The binding constraint is support time, which no source in the notes measured.

Certification and parts carry the main uncertainties. FCC estimates conflict: these notes give $800-2,000 for Part 15B alone and $2,500-5,500 in total for a product using a pre-certified module, $660 at an overseas lab, and $1,900 for an ESP32 sensor at a New Hampshire lab (a forum anecdote); the earlier field-device notes cite the same Compliance Testing page at $1,500-5,000, and SparkFun cites $4,000 for Bluetooth Part 15C testing (testing vendor) [41][149][150]. The table uses $2,500-5,500. Board prices are also moving: Raspberry Pi raised Pi 4, Pi 5 and compute-module prices with 2 GB or more of memory in February 2026 and again later in the year, leaving 1 GB models and Zero-class boards unchanged at the first round [151][152][153]; an ESP32 or Pi Zero design avoids the surcharge. Small runs cost more than they look: a solo founder's ESP32-S3 crowdfunded board cost **GBP 30.60 per sellable unit**, had to sell at $99, and lost about $18.76 per unit to campaign fees, with distributors asking 40-60% markup (self-reported) [144]; Crowd Supply notes that small batches can cost ten times more per unit than large ones (vendor) [154].

Subscription prices for comparable device businesses

The comparables split three ways: hardware subsidised by payments, device plus long contract at a high annual price, and hardware sold with a small subscription.

Company	Structure	Price or economics	Label	Source
Toast	Hardware sold at a loss; subscriptions and payments earn the margin	FY2025: subscription $936M, fintech $5,037M, hardware and professional services $180M; hardware gross profit about -$220M (third-party, unverified against the filing)	filing; aggregator	[34] [155]
Lime	Operator-owned device earns per ride	$7.47 revenue per vehicle per day; ~$1,300 landed vehicle; ~12-month payback	secondary	[62]
Flock Safety	Camera as a service, multi-year	$2,500-3,000 per camera per year plus $250-350 install; a county contract cites a $3,000 standard rate	aggregator; government contract	[33] [156] [157]
Samsara	Hardware plus 3-5 year licence	$27-45 per vehicle per month; hardware $99-148	aggregator; competitor claim	[158] [159]
Vusion	Hardware sale plus growing SaaS	Recurring services EUR 83M of EUR 1,472M (5.6%, derived); possible $0-2 per label per month cloud fee (vendor blog)	vendor	[8][5]
Cantaloupe	Monthly fee plus blended % plus equipment sold, financed or bundled	~$20 revenue per device per month (derived from secondary revenue)	filing; secondary	[16]
PayRange	Hardware plus %	BluKey $79.95; BluKey Connect $299.95 + $4.95 per location per month; 3.95% + $0.02	vendor	[13]
Nayax (resellers)	Device fee plus %	$7.95-9.99 per device per month; $30 activation	retailer	[12]
Tenant Turner	Device plus subscription	$3.50 per device per month	vendor	[32]
Trail counter data	Hardware plus data service	$35 per month per counter	search summary	[118]

The lesson is about where value pays. Device-at-location businesses with high recurring prices either sit on transaction revenue (Toast, Cantaloupe, Lime) or sell to buyers for whom the data is worth thousands a year (Flock, Samsara). A $5-10 a month subscription on a $50-100 device is at the bottom of this range, and only clears its costs if support stays near the low end of the assumption.

The customer-brings-the-hardware alternative

Selling software for hardware the customer already owns removes inventory, certification and returns. The best-known example is Nabu Casa, which says it "is profitable, has no external investors, and our only funding comes from people subscribing to Home Assistant Cloud and buying Home Assistant hardware" (vendor) [160]. Home Assistant Cloud started at $5 a month in 2018, and third-party listings put it from $6.50 (vendor) [161][162]; press reports it was renamed Home Assistant Link on 2 October 2026 [163]. Home Assistant reports more than two million households, a user count rather than a subscriber count, and the same post gives staff as 39 or 56 in different places (vendor) [164]. OctoEverywhere, which sells remote access to customer-owned 3D printers, calls itself "100% community funded" and raised its monthly price by $1 in May 2024 because of processor and service costs (vendor) [165][166]; competitor Obico charges $6.99 a month on annual billing (vendor) [167]. WLED's creator offers GitHub Sponsors tiers of $2-120 a month with no sponsor count found (self-reported) [168], and OpenEPaperLink has about 1,363 GitHub stars (aggregator) [169]. Plausible, a self-hostable analytics tool cited in the first report, received about $300 a month in donations from self-hosters (self-reported) [170].

No subscriber count or revenue figure was found for Nabu Casa, OctoEverywhere, WLED or OpenEPaperLink (no evidence found), and no small Pi or ESP32 device-plus-subscription business with disclosed sales was found; the only hardware maker with figures, Pimoroni, reported GBP 2.75M turnover in its fourth year, probably around 2016, from hardware alone (self-reported) [171]. BYOD software at $4-7 a month demonstrably survives, but on top of very large free user bases. For qrwty, BYOD is the cheaper experiment and the weaker revenue evidence at the same time.

7 Ranking and a platform view: one station, two angles

Ranking against qrwty's existing software

qrwty already issues stable redirect URLs, lets the owner change what each one does, logs each scan and runs self-hosted. The angles rank by how much of that they use, whether the device is intrinsic, and whether someone already pays.

Rank	Angle	Uses qrwty's redirect and scan log	Device intrinsic	Paying buyer evidence	Main obstacle
1	**Rotating-code presence proof** (software, any screen; station later)	Fully: each scan validates a short-lived token and the log is the proof	Yes, but any screen will do indoors	Organisers pay JPY 40,000-700,000 per campaign and venues $25-100 a month for software [28][29][30][31]; no one found paying for rotation	Price ceiling set by cheap QR/GPS tools
2	**Scan-to-actuate for mid-ticket unattended assets**	Fully: scan, Stripe Checkout, webhook, command, log	Yes: the actuator	Operators pay 3.5-6% plus device fees in vending and laundry [12][13]; no operator evidence for the mid-ticket categories	Support, failure handling, and no demand evidence
3	Print-on-event for maintenance and safety teams	Partly: server-assigned IDs and scans; redirect helps for work orders	Yes, but the printer is the customer's	$500-1,800 a year software [19][20]	Integration with record systems; incumbents in each vertical
4	Changeable-QR displays	Partly	Only for content beside the code	No SMB demand found	ESL incumbents; $749 devices; hobby tags
5	ESL proper	Marginally	No (QR is incidental)	Chains	Walmart-scale vendors

Presence proof ranks first because it uses qrwty's software most completely and costs least to test: the rotating code is a feature of the redirect server, and the display can be a tablet, an old phone or a $24 OpenEPaperLink tag the customer buys. Scan-to-actuate ranks second because the device is unambiguously intrinsic and the money flow is proven, but only below incumbents' prices on larger tickets, and with failure modes (paid but not dispensed) that create support work.

One device that serves several angles

If a device is built, one design serves angles 1, 2 and 4: an ESP32 with an e-paper panel and a relay output, on Wi-Fi with an LTE option. Its firmware does four things: pull "what to show" from the server (a rotating code, a price, a status), render it, accept signed "act" commands (pulse the relay), and report health. The server side is mostly what qrwty has or would need anyway: the redirect, a per-scan token check, the scan log, a Stripe webhook handler and a command queue. A networked label printer would be driven from the server instead of from the device, which keeps printing in the BYOD column.

Component	Presence proof	Scan-to-actuate	Changeable display	Print-on-event
Redirect with per-scan token validation	yes	yes (prevents replay)	optional	no
Scan log as an auditable event stream	yes (the proof)	yes (reconcile payments with actuations)	yes	yes
Server-pushed display content	yes (rotating code)	optional (status, price)	yes	no
Signed command to device	no	yes	no	yes (print job to OEM printer)
Payment webhook	optional (paid entries)	yes	no	no
Runs on customer-owned hardware	yes (tablet, phone, OpenEPaperLink tag)	partly (relay kit needed)	yes (OpenEPaperLink)	yes (Zebra, Brother)

What a first pilot proves

A first pilot should test the riskier of the two top angles on real hardware and the cheaper one in software. For scan-to-actuate, one operator with an unattended asset, a ticket above $20 and no incumbent platform runs 5-20 relay kits for 60-90 days. The pilot proves four things or fails: that customers complete the scan-to-pay-to-act loop at an acceptable rate, that paid-but-failed actuations stay rare and recoverable, that support stays under the assumed 2 hours per device per year, and that the operator pays a monthly fee at the end. For presence proof, one organiser with a cash-value incentive (a voucher rally, a prize draw, a paid attendance programme) uses rotating codes on its own screens and reports whether shared-photo cheating fell compared with a static-code baseline. If neither pilot produces a payment, the device thesis has no buyer.

8 Risks

Risk	Evidence	Angles most affected	Severity	Mitigation
Device swap and tamper	Fake stickers placed over genuine codes on Austin parking stations [45]; a whole device can be swapped or covered, and no official guidance recommends display codes [46]	Actuate, displays, stations	Medium	Signed device identity; rotating codes so a photographed or replaced code fails; scan anomalies flagged from the log
Quishing and user trust	Fake QR stickers on parking meters in New York (2025) and Christchurch (2026) [172][173]; the FTC tells consumers to check URLs before opening [174]; 29% of US adults in a 2024 survey named security as a QR concern (secondary) [175]	Actuate (payment surfaces)	High	Customer-owned custom domains, branded checkout, visible device identity, abuse reporting
PCI scope	Stripe Checkout supports SAQ A eligibility, but the merchant still attests annually [82][81]; the 6.4.3/11.6.1 claim is unverified [84]; Payment Links not addressed [82]	Actuate	Medium	Use hosted Checkout only; never handle card data; confirm SAQ type with the acquirer before launch
Paid but not actuated	Webhook-to-actuation brings replay, double-vend and refund risks (derived from the architecture); laundry operators report sales drops after going app-only (self-reported) [68]	Actuate	High	Idempotent commands, device acknowledgement before capture, automatic refunds, a non-app fallback
Outdoor power and connectivity	No solar or battery power costs were researched; trail counters claim up to 2-year batteries (vendor) [134]; Estimote's LTE beacon needed a data plan [135]	Stations, outdoor actuators	High	Start indoors or on mains power; e-paper holds an image with no power
Pi SD-card failure	Pi filesystems corrupt on power loss, boot read-only and stop silently [176] [177]	Any Pi-based device	Medium	Prefer ESP32; if Pi, use read-only root, eMMC or USB boot, a watchdog and A/B updates
Support burden	No support-hours or RMA benchmark found; hardware startups fail mainly from cash tied up in units (secondary) [178][179][180]	Any fleet	High	BYOD first; small paid pilots; price support explicitly
Hardware margins	Peloton hardware made 13.6% gross margin in FY2025 (secondary) [181]; Toast hardware ran at a gross loss (aggregator) [155]	Any fleet	High	Price hardware at 2.5x landed cost or leave it to the customer
Walmart-scale incumbents	Vusion EUR 1,472M revenue [8]; Cantaloupe 1.28M devices [16]; LY Corporation runs a stamp-rally product on LINE [29]	ESL, vending, Japanese rallies	High where they operate	Stay in segments they do not price for: mid-ticket independent operators, self-hosters
Japanese-market access	Rallies are bought through municipal proposal calls and specifications in Japanese [119][121]; leading products run as LINE mini apps [29][112]; no evidence on foreign vendors winning tenders	Presence proof	High for a solo non-Japanese founder	Treat Japan as price evidence, not a first market; partner with a local vendor if pursued

Evidence conflicts to resolve before relying on a number

Item	Conflict	Treatment in this report
FCC cost with a pre-certified module	$800-2,000 (15B only) and $2,500-5,500 (total) in these notes, $1,500-5,000 for the same Compliance Testing page in the earlier notes, $660 overseas, $1,900 anecdote, $4,000 for 15C [41][150]	Used $2,500-5,500
Golioth tiers	A Teams tier at $299 a month (aggregator) [182] is not on the official page, which lists $0.25 per active device per month with no tiers [35]	Used the official page
PCI 6.4.3 and 11.6.1	One vendor says they apply to hosted-checkout merchants [84]; Stripe's guide does not mention them [82]	Flagged as unconfirmed
PayRange rates	3.95% + $0.02 and $79.95 BluKey on the FAQ; 3.49% + $0.02 and $59.95 + $20 on another page [13][64]	Both shown
Noke pricing	A search snippet gave $13.90 per lock; the fetched page attributes that to another vendor [71]	Excluded
Hologram data	$0.03 per MB on the live page; $0.10 per MB on a 2023-24 page [38][183]	Used the live page
balena	Aggregators reported $2-3 per device [184]; the official page shows $159-1,439 a month plans [146]	Used the official page
Toast hardware gross profit	About -$220M from a third-party data site, not verified against the 10-K [155][34]	Labelled aggregator
Stamp-rally listing	The JPY 30,000 a month price and the JPY 5,000,000 tender cap come from one listing with uncertain attribution [111]	Labelled uncertain
Home Assistant staff	30+, 39 and 56 in different sources and within one post [164]	Not used

9 Recommendation and next experiments

Sequencing note (added 2026-10-31). The "Read this first" section of the main report places the work in this section after the paid cloud exists and has paying customers. The experiments below are specified here; their place in the overall order is decided there.

Recommendation. Do not build a fleet business yet, and do not enter ESL, regulated labelling or small-ticket vending. Build two capabilities into qrwty as software that runs on customer-owned hardware: **rotating, short-lived codes** with per-scan token validation and an optional location check, rendered on any browser screen or OpenEPaperLink tag; and a **scan-to-actuate flow** (static or rotating code, Stripe Checkout, verified webhook, signed command, acknowledgement, scan log) that drives an off-the-shelf ESP32 relay board. Add print-on-event for Zebra and Brother printers only when a maintenance or safety team asks for it. Build a qrwty reference station (ESP32, e-paper, relay) only after a pilot customer pays for the software, and price any hardware at 2.5x landed cost rather than subsidising it.

Objective for the next 90 days. Find one operator and one organiser who will pay for what only a connected device can do (act on a scan, or show a code that cannot be shared), spending under $1,000 on hardware (assumption: 10-20 relay boards at $40 [79] plus SIMs at $3 [38], or customer-owned screens).

Weeks	Experiment	Cost	Metric that decides
1-4	Ship rotating codes in qrwty: token per scan, 15-60 second window, optional geofence, a kiosk page for any browser, and an OpenEPaperLink output	Founder time	Works on a tablet and a $24 tag; screenshot older than the window fails
1-6	Ship scan-to-actuate with Stripe Checkout in test mode on a KinCony relay board, including idempotent commands and refund on failed acknowledgement	Under $100 hardware [79] [80]	1,000 simulated scans with zero double actuations and every failure refunded
3-8	Interview 10 operators of unattended assets with tickets above $20 (equipment rental, makerspace, campground amenities, EV, car wash) about current payment and access tools	Time	At least 3 have no platform or dislike theirs, and name a price
4-10	Pilot rotating codes with one organiser whose stamps or check-ins carry value (voucher, prize, paid attendance)	Time; customer screens	Organiser reports fewer invalid claims than a static baseline and asks to keep it
6-12	Paid actuation pilot: 5-20 kits at one operator for 60 days	Under $1,000 (assumption)	Operator pays a monthly fee; support under 2 hours per device per year; paid-but-failed rate under 1% (assumption)

Evidence that would change each verdict

Angle	Current verdict	Evidence that would change it
Changeable-QR displays and ESL	Not viable as ESL; display output only	SMBs paying $30-80 for hosted e-paper QR signs, or an ESL vendor asking for a QR redirect and scan-log integration
Scan-to-actuate	Conditionally viable niche	Positive: a paid pilot plus three operators naming a price. Negative: operators unwilling to move off cash or incumbents, or support above 2 hours per device per year
Event-driven printing	Software for OEM printers	A team that cannot buy a Zebra or Brother printer (outdoor, unpowered, high volume), or a vertical where no integration exists and buyers pay over $500 a year
Presence proof	Software feature first	An organiser with cash-value incentives paying for rotating codes; or one buying outdoor stations, which would justify the reference device
Fleet and BYOD	BYOD first; no subsidised fleet	A measured support rate under 0.5 hours per device per year in pilots, or any published revenue figure for BYOD device software

The founder's control-surface thesis survives in a narrower form than it was posed. The scan log as an event stream that opens, starts and proves is real and is already worth billions in vending and ticketing; the device that turns a scan into an action or a time-bound proof is the part a printed code cannot copy. What the notes do not show is a small buyer paying for that difference today, which is why the next step is two paid pilots rather than a production run.

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Glossary

Business and technical terms used in this report, in alphabetical order. Each term is linked at its first use in every section.

Term	Meaning
10-K	The annual report a US-listed company files with the SEC (Securities and Exchange Commission); audited and more reliable than press coverage.
4G / LTE	Mobile data networks; a device with a 4G or LTE modem can be online without Wi-Fi, for a monthly SIM fee.
A/B updates	Keeping two copies of a device's software so an update that fails can fall back to the previous working copy.
acquirer	The bank or payment company that processes card payments for a merchant and decides which PCI self-assessment form the merchant must file.
actuator / actuate	A part that makes something physical happen on command, such as a relay that opens a lock or starts a machine. "Scan-to-actuate" means a scan (and usually a payment) triggers that action.
ADA	Americans with Disabilities Act, the US accessibility law; relevant where a QR code is the only way to pay or enter.
Apache 2.0	A permissive open-source licence that allows use in closed products, unlike AGPL.
AWS IoT Core	Amazon's hosted message broker for connected devices; devices send and receive small messages through it.
bill of materials	The list and cost of every part in a hardware product.
BLE / Bluetooth	Short-range wireless link between a phone and a device such as a label printer. BLE (Bluetooth Low Energy) is the low-power variant.
BYOD	Bring your own device: the customer supplies the hardware (a phone, tablet, printer or hobby board) and buys only software.
Cat-1bis	A low-cost, low-speed class of LTE modem meant for connected devices rather than phones.
ChMS	Church management software; the products cited bundle attendance check-in.
crowdfunding	Pre-selling a product to many backers (for example on Crowd Supply) to pay for its first production run.
custom domain	Serving a customer's QR redirects from a web address the customer owns (for example go.theirshop.com) rather than the vendor's. Because the printed code points at their domain, they can move to another provider without reprinting.
DPP	Digital Product Passport: an EU scheme under which products carry a scannable code linking to their origin, materials and repair data.
dynamic QR code	A QR code that encodes a short redirect URL on a server rather than the final destination. The owner can change where it points and count scans; it stops working if the server does. A static QR code encodes the final content directly and can never change.
e-paper / e-ink	A display that holds its image with no power and uses energy only when the image changes; used in shelf labels and e-readers.
EBITDA	Earnings before interest, tax, depreciation and amortisation: a company's operating profit before financing and accounting charges.
eMMC	Soldered flash storage, more robust than a removable SD card.
ESL	Electronic shelf label: a small battery-powered e-paper price tag on a shop shelf, updated over a radio link.
ESP32	A cheap Wi-Fi and Bluetooth microcontroller chip (a few dollars) used in many hobby and commercial connected devices.
FCC	US Federal Communications Commission. Any device that emits radio signals must be tested and certified before sale; Part 15B covers unintentional emissions from electronics, Part 15C intentional radios such as Bluetooth.
FY	Fiscal year: a company's accounting year, which may not match the calendar year.
geofence	A virtual boundary around a place; software can refuse or accept an action depending on whether the phone is inside it.

Term	Meaning
GitHub Sponsors	GitHub's scheme for users to make recurring donations to open-source developers.
GNSS / GPS	Satellite positioning. GPS is the US system; GNSS is the general term covering GPS and the European, Russian and Chinese equivalents that phones also use.
gross margin	Revenue minus the direct cost of delivering the product (hosting, payment fees, hardware parts), as a share of revenue. Software is usually 75% or more; hardware is often far lower.
GS1 Digital Link	GS1's standard for putting a product's barcode number (its GTIN) inside a web address, so one QR code can serve both the checkout scanner and a consumer's phone.
idempotent	A command that has the same effect whether it is received once or several times, so a repeated message cannot, for example, open a lock twice or charge twice.
IFRS	International Financial Reporting Standards, the accounting rules used by listed European companies; "IFRS revenue" is the audited figure.
IP rating (IP65, IP68)	Ingress protection rating for an enclosure: IP65 means dust-tight and resistant to water jets; IP68 means it survives immersion.
JPY	Japanese yen.
kiosk	A self-service machine with a screen, such as a ticket or ordering terminal.
landed cost	The full cost of one unit delivered to the seller: parts, assembly, testing, shipping and duties, and a share of one-off costs such as certification.
LINE	Japan's dominant messaging app. "Mini apps" run inside it, so Japanese campaigns reach users without a separate download.
LOTO	Lockout/tagout: the safety procedure of locking machinery off and tagging it before maintenance.
Maximo	IBM's enterprise asset-management software, used by large maintenance organisations.
Metrc	The seed-to-sale tracking system that many US states require licensed cannabis businesses to use, with state-issued tags.
micromobility	Shared e-scooters and e-bikes rented by the minute.
NFC	Near-field communication: the tap-to-read chip technology used in contactless cards and some tags; works at a few centimetres.
OCPP	Open Charge Point Protocol: the standard by which EV chargers talk to the back-end that controls and bills them.
OEM	Original equipment manufacturer; "OEM printers" means established makers' printers (Zebra, Brother) rather than custom hardware.
open source	Software whose source code is published under a licence that lets anyone use, change and share it.
open-core	A business model where the core software is open source and free to self-host, and the company earns money from a hosted version, support or paid extras.
OTA	Over-the-air update: sending new software to devices in the field remotely.
payback period	How many months of a customer's profit it takes to recover the cost of acquiring them.
PCI DSS	Payment Card Industry Data Security Standard: the rules any business taking card payments must follow. Using a hosted checkout keeps card numbers away from the business and reduces its obligations.
quishing	Phishing by QR code: a code that leads to a fraudulent site, often a sticker placed over a genuine one.
Raspberry Pi	A small, cheap single-board computer popular for prototypes; Pi Zero is the smallest and cheapest model.
read-only root	Running a device's operating system from storage it cannot write to, so a sudden power loss cannot corrupt it.

Term	Meaning
relay	An electrically controlled switch; a relay board lets software turn a lock, motor or light on and off.
RMA	Return merchandise authorisation: a product returned under warranty; the RMA rate is the share of units that come back.
S-1	The registration document a company files with the SEC before listing its shares in the US; the first full public disclosure of its finances.
SaaS	Software as a service: software hosted by the vendor and sold by subscription.
SafeTix	Ticketmaster's mobile ticket whose barcode changes every few seconds so screenshots cannot be reused.
SAQ A	The shortest PCI DSS self-assessment questionnaire, for merchants whose payment pages are fully hosted by a provider such as Stripe. SAQ A-EP applies when the merchant's own page handles part of the payment and carries more obligations.
SD card	Removable flash memory card; the usual storage on a Raspberry Pi and a common point of failure.
self-hosting	Running the software on a server the customer controls instead of using the vendor's cloud. qrwty's open-source core allows this.
SIM	The chip or card that connects a device to a mobile network; IoT SIM providers charge a small monthly fee plus data.
SMB	Small and medium-sized business.
stamp rally	A Japanese promotion in which visitors collect a stamp at each of several locations to earn a reward; now usually done with QR codes or GPS in an app.
stars (GitHub)	A GitHub user bookmarking a project. Star counts are a rough measure of developer interest, not of use or revenue.
Stripe	A widely used online payment processor.
Stripe Checkout / Payment Links / Terminal	Stripe products: Checkout is a hosted payment page, a Payment Link is a shareable URL to one, and Terminal is Stripe's in-person card-reader service.
take rate	The percentage of each transaction a payment or marketplace platform keeps.
TLS	The encryption behind https. Each custom domain needs its own certificate, which the service must obtain and renew automatically.
token (per-scan)	A one-time or short-lived code embedded in a QR payload so that a screenshot or a replayed message is rejected.
tooling	The steel moulds needed to mass-produce a plastic enclosure; a one-off cost of thousands of dollars that only pays off over large runs.
UN 38.3	The safety test required before lithium batteries can be shipped.
VAT	Value-added tax; "ex VAT" prices exclude it.
watchdog	A hardware timer that reboots a device automatically if its software stops responding.
webhook	An automatic message one service sends to another when something happens (for example Stripe telling qrwty that a payment succeeded). A "signed" webhook carries proof that it really came from the sender.